

Linear and Sublinear Time Spectral Density Estimation

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Abstract

We analyze the popular *kernel polynomial method* (KPM) for approximating the spectral density (eigenvalue distribution) of a diagonalizable matrix $A \in \mathbb{R}^{n \times n}$ with real-valued eigenvalues. We prove that a simple and practical variant of the KPM algorithm can approximate the spectral density to ϵ accuracy in the Wasserstein-1 distance with roughly $O(\frac{1}{\epsilon})$ matrix-vector multiplications with A . This yields the first provable linear time result for the problem.

The KPM variant we study is based on *damped Chebyshev polynomial expansions*, which we also prove are stable under any approximate matrix-vector multiplication oracle. Stability allows our result to be combined with any approximation algorithm for multiplying a vector by A . As an application, we develop a randomized *sublinear* $O(n/\text{poly}(\epsilon))$ time algorithm for approximating the spectral density of an $n \times n$ normalized graph adjacency or Laplacian matrix.

Our approach leverages several tools from approximation theory, including Jackson’s seminal work on approximation with positive kernels [Jac12], and stability properties of three-term recurrence relations for orthogonal polynomials.

1 Introduction

A ubiquitous task in computational science, engineering, and data science is to extract information about the eigenvalue spectrum of a diagonalizable matrix $A \in \mathbb{R}^{n \times n}$. A full eigendecomposition takes at least $O(n^\omega)$ time¹ [Par98, BVKS19], which is prohibitively expensive for large matrices. So, we are typically interested in extracting *partial information* about the spectrum. This can be done using iterative methods like the power or Lanczos methods, which access A via a small number of matrix-vector multiplications. Each matrix-vector multiply takes at most $O(n^2)$ time to compute, and can be accelerated when A is sparse or structured, leading to fast algorithms.

However, the partial spectral information computed by such methods is limited. They typically only compute approximations to the outlying, or *largest magnitude* eigenvalues of A , missing information about the *interior* of A ’s spectrum that may be critical in applications. In network science, clusters of interior eigenvalues can be used to identify graph structures like repeated motifs [DBB19]. In deep learning, information on how the spectrum of a weight matrix differs from its random initialization can give hints about model convergence and generalization [PSG18, MM19], and Hessian eigenvalues are useful in optimization [GKX19]. Coarse information about interior eigenvalues is also used to initialize parallel GPU based methods for full eigendecomposition [AKS17, LXES19].

¹Here $\omega < 2.373$ is the fast matrix multiplication exponent.

To address these needs and many other applications, there has been substantial interest in methods for estimating the full *spectral density* of a matrix A [WWAF06]. Concretely, assume that A has all real eigenvalues $\lambda_1, \dots, \lambda_n$ and view its spectrum as a probability density s :

$$\text{Spectral density:} \quad s(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - \lambda_i). \quad (1)$$

Here δ is the Dirac delta function. The goal is to find a probability density q that approximates s in some natural metric, like the Wasserstein distance. The density q can be either continuous (represented in some closed form) or discrete (represented as a list of approximate eigenvalues $\tilde{\lambda}_1, \dots, \tilde{\lambda}_n$). See Figure 1 for an illustration. Both are useful in applications.

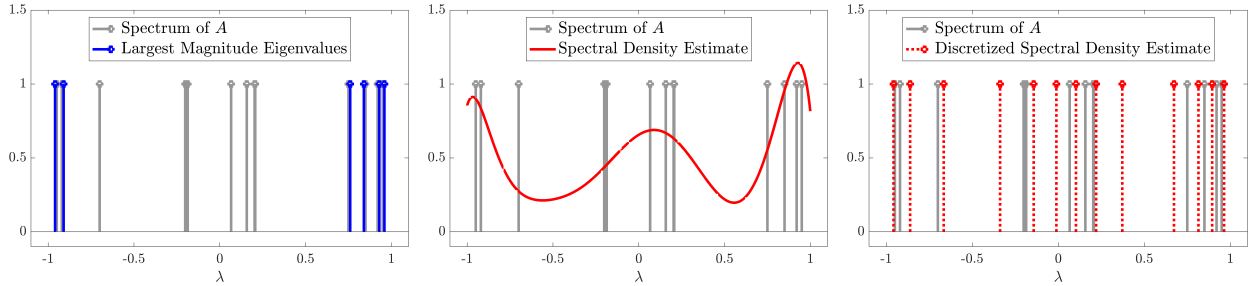


Figure 1: Different approximations for the spectrum of a symmetric matrix A , with eigenvalues in $[-1, 1]$. A typical approximation computed using an iterative eigenvalue algorithm mostly preserves information about the largest magnitude eigenvalues. In contrast, the spectral density estimates in the two right figures coarsely approximate the entire distribution of A 's eigenvalues, the first with a low-degree polynomial, and the second with a discrete distribution.

Methods for spectral density estimation that run in $o(n^\omega)$ time were first introduced for applications in condensed matter physics and quantum chemistry [Ski89, SR94, Wan94]. Many are based on the combination of two important tools: 1) moment matching, and 2) stochastic trace estimation. Specifically, if we had access to moments of the distribution s , i.e. $\frac{1}{n} \sum_{i=1}^n \lambda_i$, $\frac{1}{n} \sum_{i=1}^n \lambda_i^2$, $\frac{1}{n} \sum_{i=1}^n \lambda_i^3$, etc., then we could find a good q by finding a distribution that agrees with s on these moments. Moreover, these *spectral moments* can be computed via the matrix trace: note that $\text{tr}(A) = \sum_{i=1}^n \lambda_i$, $\text{tr}(A^2) = \sum_{i=1}^n \lambda_i^2$, $\text{tr}(A^3) = \sum_{i=1}^n \lambda_i^3$, etc. While we cannot hope to compute $\text{tr}(A^k)$ exactly in $o(n^\omega)$ time, thanks to the existence of stochastic trace estimators, like Hutchinson's method [Hut90, AT11], the trace can be approximated much more quickly. Specifically, stochastic trace estimators are based on the observation that, for any matrix $B \in \mathbb{R}^{n \times n}$, $\text{tr}(B)$ can be well approximated by $\text{tr}(G^T B G)$ where $G \in \mathbb{R}^{n \times m}$ contains random sub-Gaussian entries and $m \ll n$. For any k degree polynomial g , $G^T g(A) G$ can be computed with just $O(km)$ matrix-vector multiplications, so we can quickly approximate any low-degree moment of A 's spectral density.

This high-level approach and related techniques have been applied successfully to estimating the spectra of a wide variety of matrices [WWAF06, LSY16]. For example, the popular kernel polynomial method (KPM) is a moment matching approach, based on estimating Chebyshev or Legendre moments of A using Hutchinson's estimator. However, no worst-case approximation guarantees have been proven for such methods, making it challenging to predict their performance on general inputs and to theoretically compare different algorithmic variants.

1.1 Our contributions

The goal of this paper is to develop strong theoretical guarantees for variants of the kernel polynomial method, which is one of the most popular spectrum estimation algorithm used in practice. We target approximation in the Wasserstein-1 metric, also known as earthmover’s distance. This metric is one of the most natural for comparing a discrete distribution to a possibly continuous one (like a spectral density estimate). Given a matrix A with spectral density s , our goal is to return some representation of a density q such that:

$$W_1(s, q) \leq \epsilon. \quad (2)$$

We defer a formal definition of W_1 to Section 2. This metric has been recently used in analyzing spectral density estimation algorithms for graphs [DBB19, CKSV18]. A nice property is that, if we let $\Lambda = [\lambda_1, \dots, \lambda_n]$ be a vector of A ’s eigenvalues, then if $\tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$ satisfies $\|\Lambda - \tilde{\Lambda}\|_1 \leq n\epsilon$, we have that the discrete spectral density q with eigenvalues in $\tilde{\Lambda}$ satisfies $W_1(s, q) \leq \epsilon$.

Linear time spectral density estimation

Our first result holds for any matrix with real eigenvalues, and is stated under the assumption that these eigenvalues lie in $[-1, 1]$. It can be applied to any matrix A after scaling by $1/\|A\|_2$.

Theorem 1.1 (Linear time spectral density estimation). *Let $A \in \mathbb{R}^{n \times n}$ be a matrix with spectral density s and eigenvalues in $[-1, 1]$. Let C, C' be fixed positive constants. For any $\epsilon, \delta \in (0, 1)$, there is a variant of the kernel polynomial method (Algorithm 2, with Algorithm 3 used as a subroutine to approximate moments) which computes $T = C\ell/\epsilon$ matrix-vector multiplications with A where $\ell = \max\left(1, \frac{C'}{n}\epsilon^{-4} \log^2\left(\frac{1}{\epsilon\delta}\right)\right)$, and in $O(n\ell/\epsilon)$ additional runtime, outputs a probability density function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$ such that $W_1(s, q) \leq \epsilon$ with probability $1 - \delta$.*

Theorem 1.1 is proven in Section 5.1. Note that the result actually *improves* as n increases, requiring just $T = O(1/\epsilon)$ matrix vector multiplies when n is sufficiently large compared to $1/\epsilon$. For smaller n , the polynomial dependence on ϵ is worse. Intuitively, when A is larger, each matrix-vector product returns more information about the spectral density s , so we can estimate it more easily. The runtime of Theorem 1.1 is dominated by the cost of the matrix-vector multiplications, which take $O(T \cdot n^2)$ time to compute for a dense matrix, and $O(T \cdot \text{nnz}(A))$ time for a sparse matrix with $\text{nnz}(A)$ non-zero entries, so the algorithm runs in linear time when ϵ, δ are considered constant. The density function q returned by Algorithm 2 is in the form of a low-degree (specifically $O(1/\epsilon)$ degree) polynomial, multiplied by a simple closed form function. It is easy to plot, integrate, or manipulate. Alternatively, a simple rounding scheme due to [CKSV18] which runs in $O(n + \text{poly}(1/\epsilon))$ time can extract from q a vector of approximate eigenvalues $\tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$ satisfying $\|\Lambda - \tilde{\Lambda}\|_1 \leq 2n\epsilon$, which as discussed is 2ϵ close to the spectral density s in Wasserstein distance (see Theorem A.1).

Theorem 1.1 is based on a version of the kernel polynomial method that approximates *Chebyshev polynomial* moments: $\text{tr}(T_0(A)), \dots, \text{tr}(T_N(A))$ where $T_0, \dots, T_N$ are the Chebyshev polynomials of the first kind. We show that, if these moments are computed exactly, we can find a density function q close to s in Wasserstein-1 distance. To do so, we turn to classic work of Jackson on universal polynomial approximation bounds for Lipschitz functions: we take advantage of the fact that Jackson’s construction of such polynomials is both *linear* and *preserves positivity* [Jac12]. We

then prove that it is still possible to extract a spectral density estimation even when the Chebyshev moments are computed to relatively low precision – in contrast to methods based on the standard moments, this allows use to obtain a polynomial dependence on ϵ in Theorem 1.1.

Sublinear time spectral density estimation

Above it is assumed that our spectrum approximation algorithm can compute exact matrix-vector multiplications with A in linear time. However, in practice, spectral density estimation is often applied to matrices where we only have *implicit* access to A , and in many cases, multiplications can only be computed approximately. For example, A might be a Hessian matrix that we can multiply by approximately using Pearlmutter’s trick [Pea94, YGKM20], or the inverse of some other matrix, which we can multiply by approximately using an iterative solver. Such cases motivate our second main result, which only assumes access to an approximate matrix vector multiplication oracle:

Definition 1.2. An ϵ_{MV} -approximate matrix-vector multiplication oracle for $A \in \mathbb{R}^{n \times n}$ and error parameter $\epsilon_{MV} \in (0, 1)$ is an algorithm that, given any vector $y \in \mathbb{R}^n$, outputs a vector z such that $\|z - Ay\|_2 \leq \epsilon_{MV} \|A\|_2 \|y\|_2$. We will denote a call to such an oracle for by $AMV(A, y, \epsilon_{MV})$.

In Section 5.2 we prove the following robust version of Theorem 1.1:

Theorem 1.3 (Robust spectral density estimation). *Let $A \in \mathbb{R}^{n \times n}$ be a matrix with spectral density s and eigenvalues in $[-1, 1]$. Let C, C', C'' be fixed positive constants. For any $\epsilon, \delta \in (0, 1)$ and $\epsilon_{MV} = C''\epsilon^{-4}$, there is a variant of the kernel polynomial method (Algorithm 2, with Algorithm 4 used as a subroutine to approximate moments) which makes $T = C\ell/\epsilon$ calls to an ϵ_{MV} -approximate matrix-vector oracle for A , where $\ell = \max\left(1, \frac{C'}{n}\epsilon^{-4} \log^2\left(\frac{1}{\epsilon\delta}\right)\right)$, and in $O(n\ell/\epsilon)$ additional runtime, outputs a probability density function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$ such that $W_1(s, q) \leq \epsilon$ with probability $1 - \delta$.*

The proof of Theorem 1.3 requires showing that Chebyshev moments can be accurately computed using an approximate matrix-vector oracle. We are able to do so by drawing on stability results for the three-term recurrence relation defining these polynomials [Cle55, MMS18]. As a concrete application of Theorem 1.3, in Section 6 we show how a random sampling algorithm can be used to implement an ϵ -approximate matrix-vector multiplication oracle for any *normalized graph adjacency matrix* A in $O(n/\epsilon)$ time. Combined with Theorem 1.3, this yields:

Theorem 1.4 (Sublinear time spectral density estimation for graphs.). *Let $G = (V, E)$ be an unweighted, undirected n -vertex graph and let $A \in \mathbb{R}^{n \times n}$ be the normalized adjacency of G with spectral density s . Let $\epsilon, \delta \in (0, 1)$ be fixed values. Given access to the degrees of the vertices of G , there is an $O(n \text{poly}(\log(1/\delta)/\epsilon))$ time algorithm that outputs a density function $q : [-1, 1] \rightarrow \mathbb{R}^+$ such that $W_1(q, s) \leq \epsilon$ with probability at least $1 - \delta$.*

Since A can have up to n^2 non-zero entries, the runtime above is *sublinear* in the input size, illustrating the power of Theorem 1.3. Note that the normalized graph Laplacian $L = I - A$ has the same eigenvalues of A up to a reflection, so Theorem 1.4 also yields a sublinear time result for normalized Laplacians, whose spectral densities are of interest in network science [DBB19].

1.2 Related work

Spectral density estimation. As discussed, algorithms for spectral density estimation have been studied since the early 90s [Ski89, SR94, Wan94]. We refer the reader to the excellent survey by [WWAF06]. The Wasserstein-1 distance has been suggested as a natural metric for evaluating the error of spectral density estimates (e.g. in [DBB19]). While we are not aware of any direct results providing a linear time algorithm with a worst-case guarantee, as in Theorem 1.1, it is possible to obtain a similar result from the work of Valiant and Kong on learning a distribution in Wasserstein-1 distance based on approximations to the standard moments [KV17]. Specifically, moment estimates for A obtained via Hutchinson’s estimator could be directly plugged into that result to approximate the spectral density. However, such an approach inherently incurs an inverse *exponential* dependence on ϵ , whereas Theorem 1.1 has a polynomial dependence.

For the special case of normalized Laplacian matrices, Cohen et al. [CKSV18] prove a result which matches the guarantee of Theorem 1.4, but with runtime of $2^{O(1/\epsilon)}$ – i.e., with *no dependence* on n . In comparison, our result depends linearly on n , but only polynomially on $1/\epsilon$. An interesting open question is if a $1/\text{poly}(\epsilon)$ time algorithm is possible.

Matrix-vector query algorithms. Our work fits into a broader line of work on proving upper and lower bounds on the *matrix-vector query complexity* of linear algebraic problems, from top eigenvector, to matrix inversion, to rank estimation [SWYZ19, SEAR18, BHSW20, MMMW20, DM21]. The goal in this model is to minimize the total number of matrix-vector multiplications with A , recognizing that such multiplications either 1) dominate runtime cost or 2) are the only way to access A when it is an implicit matrix. The matrix-vector query model generalizes both classical Krylov subspace methods, as well as randomized sketching methods [Woo14]. Studying other basic linear algebra problem when matrix-vector multiplication queries are only assumed to be approximate (as in Definition 1.2) is an interesting future direction.

1.3 Paper Roadmap

We describe notation and preliminaries on polynomial approximation in Section 2. We use these tools in Section 3 and Section 4 to prove that a good approximation to the first $O(1/\epsilon)$ Chebyshev polynomial moments of the spectral density can be used to extract a good approximation in Wasserstein-1 distance. This result is the basis for our main results, Theorem 1.1 and Theorem 1.3, which are proven in Section 5. Finally, we prove Theorem 1.4 in Section 6.

2 Preliminaries

Throughout we assume that $A \in \mathbb{R}^{n \times n}$ is diagonalizable with eigendecomposition $A = U\Lambda U^{-1}$ and real eigenvalues $-1 \leq \lambda_n \leq \dots \leq \lambda_1 \leq 1$. Spectral density estimation is often applied to symmetric matrices, in which case the eigendecomposition is orthogonal: $A = U\Lambda U^T$. We denote A ’s spectral density by s , which is defined in (1). Our goal is to approximate s in the Wasserstein-1 metric with another distribution q supported on $[-1, 1]$. Specifically, as per the dual formulation given by the

Kantorovich-Rubinstein theorem [KR57], for s, q supported on $[-1, 1]$ the metric is equal to:

$$W_1(s, q) = \sup_{\substack{f: \mathbb{R} \rightarrow \mathbb{R} \\ |f(x) - f(y)| \leq |x - y| \quad \forall x, y}} \left\{ \int_{-1}^1 f(x) (s(x) - q(x)) dx \right\}. \quad (3)$$

In words, s and q are close in Wasserstein-1 distance if their difference has small inner product with all 1-Lipschitz functions f . Alternatively, $W_1(s, q)$ is equal to the cost of “changing” one distribution to another, where the cost of moving one unit of mass from x to y is $|x - y|$: this is the “earthmover’s” formulation common in computer science. Note that (3) can be applied to arbitrary functions s, q , even if they are *not distributions*, and we will occasionally do so.

Functions and inner products. We introduce notation for functions used throughout the paper. Let $\mathcal{F}([-1, 1], \mathbb{R})$ denote the space of real-valued functions on $[-1, 1]$. For $g, h \in \mathcal{F}([-1, 1], \mathbb{R})$, let $\langle g, h \rangle$ denote $\langle g, h \rangle := \int_{-1}^1 g(x)h(x)dx$. For $f \in \mathcal{F}([-1, 1], \mathbb{R})$, we define $\|f\|_2 := \sqrt{\langle f, f \rangle}$ and let $\|f\|_\infty$ denote the max-norm $\|f\|_\infty = \max_{x \in [-1, 1]} |f(x)|$. We let $\|f\|_1$ denote $\|f\|_1 = \int_{-1}^1 |f(x)|dx$.

Let $\mathcal{F}(\mathbb{Z}, \mathbb{R})$ be the space of real-valued functions on the integers, $\mathbb{Z}$. For $f, g \in \mathcal{F}(\mathbb{Z}, \mathbb{R})$ let $(f * g)$ denote the discrete convolution: $(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m]g[n - m]$. Let $\mathcal{F}(\mathbb{N}, \mathbb{R})$ be the space of real-valued functions on the natural numbers, $\mathbb{N}$. For functions in $\mathcal{F}(\mathbb{Z}, \mathbb{R})$ or $\mathcal{F}(\mathbb{N}, \mathbb{R})$ we typically used square brackets instead of parentheses.

For two functions f, g let $h = fg$ (or $h = f \cdot g$) and $j = f/g$ denote the pointwise product and quotient respectively. I.e. $h(x) = f(x)g(x)$ and $j(x) = f(x)/g(x)$ for all x .

Chebyshev polynomials. Our approach is based on approximating Chebyshev polynomial moments of A ’s spectral density, and we will use basic properties of these polynomials, the k^{th} of which we denote T_k . The Chebyshev polynomial of the first kind can be defined via the recurrence:

$$\begin{aligned} T_0(x) &= 1 & T_1(x) &= x \\ T_k(x) &= 2x \cdot T_{k-1}(x) - T_{k-2}(x) & \text{for } k \geq 2. \end{aligned}$$

We will use the well known fact that the Chebyshev polynomials of the first kind are bounded between $[-1, 1]$, i.e. $\max_{x \in [-1, 1]} |T_k(x)| \leq 1$.

Let $w(x) := \frac{1}{\sqrt{1-x^2}}$. It is well known that $\langle T_0, w \cdot T_0 \rangle = \pi$, $\langle T_k, w \cdot T_k \rangle = \pi/2$ for $k > 0$, and

$$\langle T_i, w \cdot T_j \rangle = 0 \quad \text{for } i \neq j.$$

In other words, the Chebyshev polynomials are orthogonal on $[-1, 1]$ under the weight function w . The first k Chebyshev polynomials form an orthogonal basis for the degree k polynomials under this weight function. We let $\bar{T}_k$ denote the *normalized* Chebyshev polynomial $\bar{T}_k := T_k / \sqrt{\langle T_k, w \cdot T_k \rangle}$.

Definition 2.1 (Chebyshev Series). The *Chebyshev expansion or series* for a function $f \in \mathcal{F}([-1, 1], \mathbb{R})$ is given by

$$\sum_{k=0}^{\infty} \langle f, w \cdot \bar{T}_k \rangle \cdot \bar{T}_k.$$

We call $\sum_{k=0}^N \langle f, w \cdot \bar{T}_k \rangle \cdot \bar{T}_k$ the truncated Chebyshev expansion or series of degree N .

Other notation. Let $[n]$ denote $1, \dots, n$. For a scalar function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $n \times n$ matrix A with eigendecomposition $U\Lambda U^{-1}$, we let $f(A)$ denote the matrix function $Uf(\Lambda)U^{-1}$. Here $f(\Lambda)$ is understood to mean f applied entrywise to the diagonal matrix Λ containing A 's eigenvalues. Note that $\text{tr}(f(A)) = \sum_{i=1}^n f(\lambda_i)$. When $f(x)$ is a degree q polynomial, $c_0 + c_1x + \dots, c_qx^q$, then we can check that $f(A)$ exactly equals $c_0I + c_1A + \dots, c_qA^q$, where I is then $n \times n$ identity matrix. So $f(A)y$ can be computed for any vector y using q matrix-vector multiplications with A .

3 Idealized Kernel Polynomial Method

In this section we show that the spectral density s of a matrix A with eigenvalues in $[-1, 1]$ can be well approximated given access to the first $N = O(1/\epsilon)$ Chebyshev polynomial moments of s , i.e., to $\text{tr}(\bar{T}_0(A)), \dots, \text{tr}(\bar{T}_N(A))$. This result will form the basis for the full kernel polynomial method we analyze in Section 4, which approximates these moments using a stochastic trace estimator, implemented with either exact or approximate matrix vector multiplications with A .

A natural approach is to use these moments to construct a truncated Chebyshev series approximation to s (see Definition 2.1). To do so, note that the scaled moments $\frac{1}{n} \text{tr}(\bar{T}_0(A)), \dots, \frac{1}{n} \text{tr}(\bar{T}_N(A))$ are exactly equal to the first N Chebyshev series coefficients of s/w , where $w(x) = \frac{1}{\sqrt{1-x^2}}$ is as defined in Section 2. Specifically, the eigenvalues of $\bar{T}_k(A)$ are equal to $\bar{T}_k(\lambda_1), \dots, \bar{T}_k(\lambda_n)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . Since the trace of a diagonalizable matrix is the sum of its eigenvalues, we have $\frac{1}{n} \text{tr}(\bar{T}_k(A)) = \frac{1}{n} \sum_{i=1}^n \bar{T}_k(\lambda_i) = \langle s, \bar{T}_k \rangle = \langle s/w, w \cdot \bar{T}_k \rangle$.

After using the scaled Chebyshev moments to construct a truncated Chebyshev series for s/w , i.e. a degree N polynomial approximation, we can then multiply the final result by w to obtain an approximation to s . Unfortunately, there are two issues with this approach: 1) it is difficult to analyze the quality of the Chebyshev series approximation, since s is not a smooth function, and 2) this approximation will not in general be a non-negative function, which is a challenge because our goal is to find *probability density* that well approximates s .

A common approach for dealing with the second issue is to instead use a *damped Chebyshev expansion* [WWAF06], where the Chebyshev coefficients are slightly reweighted to ensure that the resulting polynomial is always non-negative. Such non-negativity preserving damping schemes follow from the connection between Chebyshev and Fourier series: we refer the reader to Appendix B for details. In short, by the convolution theorem, Fourier series truncation corresponds to convolution with a function whose Fourier support is bounded.

If this function is also non-negative, convolution preserves non-negativity of the function being approximated, leading to truncated series that is guaranteed to be positive. One such damping schemes was introduced in classic work of Jackson [Jac12]. For any positive integer z , let $N = 4z$. Then, for $k = 0, \dots, N$, define the coefficient

$$\hat{b}_N[k] = \sum_{j=-\frac{N}{2}-1}^{\frac{N}{2}+1-k} \left(\frac{N}{2} + 1 - |j| \right) \cdot \left(\frac{N}{2} + 1 - |j+k| \right). \quad (4)$$

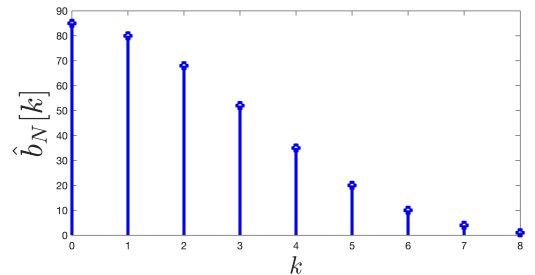


Figure 2: Jackson coefficients for $N = 8$.

While (4) may look opaque, $\hat{b}_N[0], \dots, \hat{b}_N[N]$ are actually equal to the result of a simple discrete convolution operation. Let $g \in \mathcal{F}(\mathbb{Z}, \mathbb{R})$ have $g[j] = 1$ for $j = -z, \dots, z$, and $g[j] = 0$ otherwise. Then let $\hat{b}_N = (g * g) * (g * g)$ and $\hat{b}_N[0], \dots, \hat{b}_N[N]$ be the values corresponding to non-negative indices.² See Fig. 2 for an illustration of these coefficients. They are all positive and $\hat{b}_N[0] > \hat{b}_N[1] > \dots > \hat{b}_N[N]$. Jackson suggests approximating a function using the following truncation based on these coefficients:

Definition 3.1 (Jackson damped Chebyshev series). Let $f \in \mathcal{F}([-1, 1], \mathbb{R})$ have Chebyshev series $\sum_{k=0}^{\infty} \langle f, w \cdot \bar{T}_k \rangle \cdot \bar{T}_k$. The Jackson approximation to f is a degree N polynomial $\bar{f}_N$ obtained via the following truncation with modified coefficients:

$$\bar{f}_N(x) := \sum_{k=0}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \langle f, w \cdot \bar{T}_k \rangle \bar{T}_k(x). \quad (5)$$

Note that $\hat{b}_N[0]/\hat{b}_N[0] = 1$, and all other terms are strictly less than one. It is not hard to show this damped series preserves positivity. We prove the following fact as Lemma B.7 in the appendix:

Fact 3.2. *If $f : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$ is a non-negative function, then the polynomial $\bar{f}_N(x)$ defined in (5) is non-negative for all $x \in [-1, 1]$.*

Beyond preserving non-negativity, the Jackson damped Chebyshev approximation is more well-known for the fact that it provably provides a good *uniform* polynomial approximation to any Lipschitz function. In fact, an analysis of (5) is how Jackson obtained his famous result on the approximation of Lipschitz functions [Jac30]. Concretely, we have:

Fact 3.3. *Let $f \in \mathcal{F}([-1, 1], \mathbb{R})$ be a Lipschitz continuous function with Lipschitz constant λ . I.e., $|f(x) - f(y)| \leq \lambda|x - y|$ for all $x, y \in [-1, 1]$. For $N \in 4\mathbb{N}^+$, let $\bar{f}_N$ be the degree N polynomial approximation to f from (5). This approximation satisfies $\|\bar{f}_N - f\|_{\infty} \leq 18 \frac{\lambda}{N}$.*

For completeness, we give a proof of this fact as Theorem B.6 in the appendix. With Facts 3.2 and 3.3 in place, we are ready to introduce the basic kernel polynomial method for approximating the spectral density s – it is identical to the “Jackson Kernel” KPM from [WWAF06]. Recall that, for now, we assume we have access to *exact* Chebyshev moment of the spectral density s for our matrix A . In Section 4 we prove that Algorithm 1 is robust to using approximate moments.

Algorithm 1 Idealized Jackson Damped Kernel Polynomial Method

Input: Diagonalizable $A \in \mathbb{R}^{n \times n}$ with spectral density $s : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$, degree $N \in 4\mathbb{N}^+$.

Output: Density function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$.

- 1: For $k = 0, \dots, N$ compute $\tau_k = \frac{1}{n} \text{tr}(\bar{T}_k(A)) = \langle s, \bar{T}_k \rangle$.
 - 2: For $k = 0, \dots, N$ compute $\hat{b}_N[k]$ as is (4).
 - 3: Return $q = w \cdot \sum_{k=0}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \cdot \tau_k \cdot \bar{T}_k$.
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Lemma 3.4. *If $N \geq \frac{18}{\epsilon}$, then the function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$ returned by Algorithm 1 is a probability density and satisfies:*

$$W_1(s, q) \leq \epsilon.$$

²This formulation allows the coefficients to be easily computed in most high-level programming languages. E.g., in MATLAB we can compute `g = ones(2*z+1,1); c = conv(conv(g,g),conv(g,g)); b = c(N+1:2*N+1);`.

Proof. We first prove that q is a probability density. To see that it is positive, note that $h = \sum_{k=0}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \cdot \tau_k \cdot \bar{T}_k$ is a Jackson approximation to the positive function s/w , so it must be non-negative by Fact 3.2. Since w is also non-negative, we conclude that $q = w \cdot h$ is as well. Then we consider q 's integral. We need to show that $\int_{-1}^1 q(x)dx = 1 = \int_{-1}^1 s(x)dx$. Since $\bar{T}_0$ is a scaling of the constant function, it suffices to show that $\langle \bar{T}_0, q \rangle = \langle \bar{T}_0, s \rangle$. We have:

$$\langle \bar{T}_0, q \rangle = \tau_0 \cdot \langle \bar{T}_0, w \cdot T_0 \rangle = \langle \bar{T}_0, s \rangle \cdot 1.$$

The first step follows directly from the orthogonality of the Chebyshev polynomials under the weight function w , which implies that $\langle \bar{T}_0, w \cdot T_k \rangle = 0$ for all $k > 0$. We also use that $T_0 = \langle \bar{T}_0 = 1$.

Next, we prove the approximation guarantee. Referring to the formulation of Wasserstein-1 distance from equation (3), we have that $W_1(s, q) = \sup \langle s - q, f \rangle$ where f is a 1-Lipschitz function. So, we want to show that any 1-Lipschitz f has small inner product with the difference between s and its degree- N Jackson approximation, q . To do so, we show that this inner product is actually *exactly equal* to the inner product between s and a degree- N Jackson approximation to f . Since f is 1-Lipschitz, this approximation is guaranteed to have small error. This key equivalency follows because, like a standard Chebyshev series approximation, the Jackson approximation can be viewed as the output of a symmetric linear operator applied to s .

Formally, we introduce notation for several linear operators needed to analyze (5). Let $\bar{\mathcal{T}} : \mathcal{F}([-1, 1], \mathbb{R}) \rightarrow \mathcal{F}(\mathbb{N}, \mathbb{R})$ be the operator mapping a function $f \in \mathcal{F}([-1, 1], \mathbb{R})$ to its inner-product with the normalized Chebyshev polynomials. Define the transpose operator $\bar{\mathcal{T}}^* : \mathcal{F}(\mathbb{N}, \mathbb{R}) \rightarrow \mathcal{F}([-1, 1], \mathbb{R})$ to satisfy $\langle \bar{\mathcal{T}}f, g \rangle = \langle f, \bar{\mathcal{T}}^*g \rangle$ for any $g \in \mathcal{F}(\mathbb{N}, \mathbb{R})$. Concretely, for $i \in \mathbb{N}$ and $x \in [-1, 1]$,

$$[\bar{\mathcal{T}}f][i] := \int_{-1}^1 \bar{T}_i(x)f(x)dx \quad \text{and} \quad [\bar{\mathcal{T}}^*g](x) := \sum_{i=0}^{\infty} \bar{T}_i(x)g[i]. \quad (6)$$

We also define operators $\mathcal{W} : \mathcal{F}([-1, 1], \mathbb{R}) \rightarrow \mathcal{F}([-1, 1], \mathbb{R})$ and $\mathcal{I} : \mathcal{F}(\mathbb{N}, \mathbb{R}) \rightarrow \mathcal{F}(\mathbb{N}, \mathbb{R})$ as follows:

$$[\mathcal{W}f](x) := w(x)f(x) = \frac{1}{\sqrt{1-x^2}}f(x) \quad \text{and} \quad [\mathcal{I}g][i] := g[i].$$

Note that $\mathcal{I}$ is an identity operator. For any $N \in 4\mathbb{N}$, we define $\mathcal{B}_N : \mathcal{F}(\mathbb{N}, \mathbb{R}) \rightarrow \mathcal{F}(\mathbb{N}, \mathbb{R})$ as:

$$[\mathcal{B}_N g](i) := \begin{cases} \frac{\hat{\beta}_N[i]}{\hat{\beta}_N[0]}g(i) & \text{for } 0 \leq i \leq N \\ 0 & i > N. \end{cases}$$

The operators $\mathcal{W}$, $\mathcal{I}$, and $\mathcal{B}_N$ are all commutative with respect to the inner-products in their respective spaces. Specifically, for $f_1, f_2 \in \mathcal{F}([-1, 1], \mathbb{R})$ and $g_1, g_2 \in \mathcal{F}(\mathbb{N}, \mathbb{R})$, $\langle f_1, \mathcal{W}f_2 \rangle = \langle \mathcal{W}f_1, f_2 \rangle$, $\langle g_1, \mathcal{I}g_2 \rangle = \langle \mathcal{I}g_1, g_2 \rangle$, and $\langle g_1, \mathcal{B}_N g_2 \rangle = \langle \mathcal{B}_N g_1, g_2 \rangle$. Also note that by orthogonality of the Chebyshev polynomials under w , $\bar{\mathcal{T}}^* \bar{\mathcal{T}} \mathcal{W}$ is the identity operator on $\mathcal{F}([-1, 1], \mathbb{R})$ and so is $\mathcal{W} \bar{\mathcal{T}}^* \bar{\mathcal{T}}$.

With these operators defined, the remainder of the proof is short. We have via direct calculation:

$$\begin{aligned} \langle f, s - q \rangle &= \langle f, s - \mathcal{W} \bar{\mathcal{T}}^* \mathcal{B}_N \bar{\mathcal{T}} s \rangle \\ &= \langle f, \mathcal{W} \bar{\mathcal{T}}^* (\mathcal{I} - \mathcal{B}_N) \bar{\mathcal{T}} s \rangle = \langle \bar{\mathcal{T}}^* (\mathcal{I} - \mathcal{B}_N) \bar{\mathcal{T}} \mathcal{W} f, s \rangle = \langle f - \bar{\mathcal{T}}^* \mathcal{B}_N \bar{\mathcal{T}} \mathcal{W} f, s \rangle. \end{aligned}$$

Note that $\bar{\mathcal{T}}^* \mathcal{B}_N \bar{\mathcal{T}} \mathcal{W} f$ is exactly the degree- N Jackson approximation to f . So by Fact 3.3, if f is a 1-Lipschitz function, $\|f - \bar{\mathcal{T}}^* \mathcal{B}_N \bar{\mathcal{T}} \mathcal{W} f\|_{\infty} \leq 18/N$. Since s is a non-negative function that integrates to 1, it follows that $\langle f, s - q \rangle = \langle f - \bar{\mathcal{T}}^* \mathcal{B}_N \bar{\mathcal{T}} \mathcal{W} f, s \rangle \leq 18/N$. Since $W(s, q) = \sup_{1\text{-Lipschitz } f} \langle f, s - q \rangle$, we conclude that $W(s, q) \leq \epsilon$ as long as $N \geq 18/\epsilon$. $\square$

Remark. Given access to the Chebyshev polynomial moments, $\text{tr}(\bar{T}_0(A)), \dots, \text{tr}(\bar{T}_N(A))$, Algorithm 1 can be implemented in $O(1/\epsilon)$ additional time. The function it returns is an $O(1/\epsilon)$ degree polynomial times the closed form function w . The polynomial can be represented as a sum of Chebyshev polynomials, or converted to standard monomial form in $O(1/\epsilon^2)$ time. The function is easily plotted or integrated over a range – see discussion around Fact A.2 for more details.

4 Full Kernel Polynomial Method

In reality, it is not possible to efficiently compute the exact Chebyshev polynomial moments, $\text{tr}(\bar{T}_0(A)), \dots, \text{tr}(\bar{T}_N(A))$. To obtain the linear and sublinear time results of Theorems 1.1 and Theorems 1.4, we need to work with approximations to these moments, which can be computed using a stochastic trace estimator. We introduce the details of how to do so in Section 5. Here, we first prove a general result on the accuracy of approximation needed to ensure we obtain a good spectral density estimation. Specifically, we analyze the following “robust” version of Algorithm 1.

Algorithm 2 Jackson Damped Kernel Polynomial Method

Input: Diagonalizable $A \in \mathbb{R}^{n \times n}$ with spectral density $s : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$, degree parameter $N \in 4\mathbb{N}^+$, algorithm $\mathcal{M}(A)$ that computes moment approximations $\tilde{\tau}_1, \dots, \tilde{\tau}_N$ with the guarantee that $|\tilde{\tau}_k - \frac{1}{n} \text{tr}(\bar{T}_k(A))| \leq 1/N^2$ for all k .

Output: Density function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$.

- 1: For $k = 1, \dots, N$ use $\mathcal{M}$ to compute $\tilde{\tau}_1, \dots, \tilde{\tau}_N$ as above. Set $\tilde{\tau}_0 = 1/\sqrt{\pi}$.
 - 2: For $k = 0, \dots, N$ compute $\hat{b}_N[k]$ as is (4).
 - 3: Compute polynomial $\tilde{s}_N = w \cdot \sum_{k=0}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \cdot \tilde{\tau}_k \cdot \bar{T}_k$.
 - 4: Return the probability density $q = \left(\tilde{s}_N + \frac{w\sqrt{2}}{N\sqrt{\pi}} \right) / \left(1 + \frac{\sqrt{2\pi}}{N} \right)$.
-

The final transformation of $\tilde{s}_N$ in Line 4 of Algorithm 2 ensures that we return a proper density, since error incurred by approximating $\frac{1}{n} \text{tr}(\bar{T}_k(A)) = \langle s, \bar{T}_k \rangle$ could leave the function with negative values. So, we shift by a small positive function, and rescale to maintain unit integral. Our main result on the error of Algorithm 2, which parallels Lemma 3.4 for Algorithm 1, is as follows:

Lemma 4.1. *If $N \geq \frac{18}{\epsilon}$, then the function $q : [-1, 1] \rightarrow \mathbb{R}^{\geq 0}$ returned by Algorithm 2 is a probability density and satisfies:*

$$W_1(s, q) \leq 2\epsilon.$$

Remark. When $N = O(1/\epsilon)$, as required by Lemma 4.1, Algorithm 2 only requires a $1/N^2 = O(\epsilon^2)$ approximation to the Chebyshev polynomial moments, which is rather coarse in that it depends polynomially, rather than exponentially, on ϵ . As we will see in the next section, such an approximation is easily obtained with $\text{poly}(1/\epsilon)$ matrix-vector multiplications with A , and for some matrices, can even be obtained with a sublinear time algorithm.

Proof. We first prove that q is a probability distribution. Let s_N denote the ideal distribution returned by Algorithm 1 if exact Chebyshev moments were used. I.e.,

$$s_N = w \cdot \sum_{k=0}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \cdot \tau_k \cdot \bar{T}_k$$

where $\tau_k = \frac{1}{n} \text{tr}(\bar{T}_k(A)) = \langle s, \bar{T}_k \rangle$. Note that for any density s , $\tau_0 = \langle s, \bar{T}_0 \rangle = 1/\sqrt{\pi}$. Let $\Delta_k = \tilde{\tau}_k - \tau_k$. We have $\tilde{s}_N(x) = s_N + \sum_{k=1}^N \Delta_k \frac{\hat{b}_N[k]}{\hat{b}_N[0]} w(x) \bar{T}_k(x)$. Define functions $\eta = s_N/w$ and $\tilde{\eta} = \tilde{s}/w$. It follows that for any $x \in [-1, 1]$,

$$|\tilde{\eta}(x) - \eta(x)| = \left| \sum_{k=1}^N \frac{\hat{b}_N[k]}{\hat{b}_N[0]} \Delta_k \bar{T}_k(x) \right| \leq \frac{\sqrt{2}}{N\sqrt{\pi}}. \quad (7)$$

The last inequality uses that $0 \leq \hat{b}_N[k]/\hat{b}_N[0] \leq 1$ and for $x \in [-1, 1]$, $\bar{T}_k(x) \leq \sqrt{2/\pi}$ for $k \geq 1$. Since η is a non-negative function, from (7) we can conclude that the function $\tilde{\eta} + \frac{\sqrt{2}}{N\sqrt{\pi}}$ is non-negative, and thus $w \cdot (\tilde{\eta} + \frac{\sqrt{2}}{N\sqrt{\pi}}) = \tilde{s}_N + w \frac{\sqrt{2}}{N\sqrt{\pi}}$ is also non-negative. The density of this function is $\int_{-1}^1 \tilde{s}_N(x) dx + \frac{\sqrt{2}}{N\sqrt{\pi}} \int_{-1}^1 w(x) dx = 1 + \frac{\sqrt{2\pi}}{N}$, so dividing by $1 + \frac{\sqrt{2\pi}}{N}$ gives a probability density.

Next we prove the approximation guarantee. By Lemma 3.4 we know that $W_1(s, s_N) \leq \epsilon$, so if we can show that $W_1(s_N, q) \leq \epsilon$, then by triangle inequality we will have shown that $W_1(s, q) \leq W_1(s, s_N) + W_1(s_N, q) \leq 2\epsilon$.

To bound $W_1(s_N, q)$, we need to show that $\langle f, s_N - q \rangle \leq \epsilon$ for any 1-Lipschitz function $f \in \mathcal{F}([-1, 1], \mathbb{R})$. Without loss of generality, we can assume that $\int_{-1}^1 f(x) dx = 0$, as the 1-Lipschitz function $f' = f - \int_{-1}^1 f(x) dx$ satisfies $\langle f, s_N - q \rangle = \langle f', s_N - q \rangle$ (since s_N and q are both probability densities). If $\int_{-1}^1 f(x) dx = 0$, $f(x)$ must be zero for some $x \in [-1, 1]$, and since it is also 1-Lipschitz we can in turn bound $\|f\|_\infty \leq 1$.³ We can then bound the inner product:

$$\begin{aligned} \langle f, s_N - q \rangle &\leq \|f(\tilde{s}_N - q)\|_1 \leq \|f\|_\infty \|\tilde{s}_N(x) - q(x)\|_1 \leq \left\| w \cdot \left(\eta - \frac{\tilde{\eta} + \frac{\sqrt{2}}{N\sqrt{\pi}}}{1 + \frac{\sqrt{2\pi}}{N}} \right) \right\|_1 \\ &\leq \underbrace{\left\| w \cdot \left(\eta - \tilde{\eta} - \frac{\sqrt{2}}{N\sqrt{\pi}} \right) \right\|_1}_{z_1} + \underbrace{\left\| \frac{\sqrt{2\pi}}{N} \cdot w \cdot \left(\tilde{\eta} + \frac{\sqrt{2}}{N\sqrt{\pi}} \right) \right\|_1}_{z_2} \end{aligned}$$

The last inequality uses the fact that $1 - \frac{1}{1+\gamma} \leq \gamma$ for $0 \leq \gamma \leq 1$, which we apply with $\gamma = \frac{\sqrt{2\pi}}{N}$. Using the fact that $\|w\|_1 = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \pi$ and the bound on $\|\eta - \tilde{\eta}\|_\infty$ from (7), we have

$$z_1 \leq \|w\|_1 \cdot \left\| \eta - \tilde{\eta} - \frac{\sqrt{2}}{N\sqrt{\pi}} \right\|_\infty \leq \frac{2\pi\sqrt{2}}{N\sqrt{\pi}}.$$

Examining z_2 , recall that we showed earlier that $w(\tilde{\eta} + \frac{\sqrt{2}}{N\sqrt{\pi}}) = \tilde{s}_N + \frac{\sqrt{2}}{N\sqrt{\pi}} w$ has ℓ_1 norm $1 + \frac{\sqrt{2\pi}}{N}$. So we have $z_2 \leq \frac{\sqrt{2\pi}}{N} (1 + \frac{\sqrt{2\pi}}{N}) \leq \frac{2\sqrt{2\pi}}{N}$ for all $N \geq 1$.

Compiling the bounds on z_1 and z_2 , we have that for all 1-Lipschitz f , $\langle f, s_N - q \rangle \leq \frac{4\sqrt{2\pi}}{N} \leq \frac{11}{N}$, and thus $W_1(\tilde{s}_N, q) \leq \frac{11}{N}$. For $N \geq \frac{18}{\epsilon}$ we conclude that $W_1(\tilde{s}_N, q) \leq \epsilon$. Applying triangle quality as discussed above completes the proof. $\square$

³Let z maximize $f(x)$. Since f is 1-Lipschitz we have $f(z) \leq |x - z| - f(x)$ for all x . Integrating both sides from -1 to 1 , we have $2f(z) \leq (z^2 + 1) - 0 \leq 2$. So, $f(z) \leq 1$.

5 Efficient Chebyshev Moment Approximation

With Lemma 4.1 in place, we are ready to prove our main results. To do so, we need to show how to efficiently approximate the first N Chebyshev moments of a matrix A 's spectral density s , as required by Algorithm 2. Recall that the k^{th} normalized Chebyshev moment of s is equal to $\langle s, \bar{T}_k \rangle = \frac{1}{n} \text{tr}(\bar{T}_k(A))$. We will prove that this trace can be approximated using Hutchinson's stochastic trace estimator, implemented with either exact or approximate matrix-vector multiplications with A .

This estimator requires repeatedly computing $\bar{T}_k(A)g$ for a random vector g , which is done using the standard three-term (forward) recurrence for the Chebyshev polynomials and requires a total of k matrix-vector multiplications with A . We analyze the basis approach in Section 5.1, which yields Theorem 1.1. Then in Section 5.2, we argue that the approach is stable even when implemented with approximate matrix-vector multiplication, which yields Theorem 1.3.

5.1 Exact Matrix-Vector Multiplications

Hutchinson's estimator is a widely used estimator to efficiently compute accurate estimates of $\text{tr}(R)$ for any square matrix $R \in \mathbb{R}^{n \times n}$. Each instance of the estimator computes the quadratic form $g^\top R g$ for a random vector $g \in \{-1, 1\}^n$ whose entries are Rademacher random variables. This is an unbiased estimator for $\text{tr}(R)$ with variance $\leq 2\|R\|_F^2$, and its error has been analyzed in several earlier results [AT11, RA15]. We apply a standard high-probability bound from [MMM20, RV13]:

Lemma 5.1 (Lemma 2, [MMM20]).⁴ *Let $R \in \mathbb{R}^{n \times n}$, $\delta \in (0, 1/2]$, $l \in \mathbb{N}$. Let $g^{(1)}, \dots, g^{(l)} \in \{-1, 1\}^{n \times n}$ be ℓ random vectors with i.i.d $\{-1, +1\}$ random entries. For a fixed constant C , with probability at least $1 - \delta$,*

$$\left| \text{tr}(R) - \frac{1}{\ell} \sum_{i=1}^l (g^{(i)})^\top R g^{(i)} \right| \leq \frac{C \log(1/\delta)}{\sqrt{\ell}} \|R\|_F.$$

For a polynomial $p \in \mathcal{F}([-1, 1], \mathbb{R})$ with degree k , applying Hutchinson's estimator to $R = p(A)$ requires computing $p(A)g$, which can always be done with k matrix-vector multiplies with A . If $p(x)$ admits a recursive construction, like the Chebyshev polynomials, then this recurrence can be used. Specifically, for the Chebyshev polynomials, we have:

$$\begin{aligned} T_0(A)g &= g & T_1(A)g &= Ag \\ T_k(A)g &= 2A \cdot T_{k-1}(A)g - T_{k-2}(A)g & \text{for } k \geq 2. \end{aligned} \tag{8}$$

A moment estimation algorithm based on Hutchinson's estimator is stated as Algorithm 3.

Remark. In total, Algorithm 3 requires $N \cdot \ell$ matrix multiplications with A since for each i $T_1(A)g^{(i)}, \dots, T_N(A)g^{(i)}$ can be computed using the same N steps of the (8) recurrence. It requires $O(n\ell N)$ additional runtime to compute and sum all inner products of the form $(g^{(i)})^\top T_k(A)g^{(i)}$.

Our main bound on the accuracy of Algorithm 3 follows:

⁴In [MMM20] the lemma is stated with an assumption that $\ell > O(1/\delta)$. However, it is easy to see that the same claim holds without this assumption, albeit with a quadratically worse $\log(1/\delta)$ dependence. The proof follows from same application of the Hanson-Wright inequality used in that work.

Algorithm 3 Hutchinson Moment Estimator

Input: Diagonalizable $A \in \mathbb{R}^{n \times n}$ with $\|A\|_2 \leq 1$, degree $N \in 4\mathbb{N}^+$, number of repetitions $\ell \in \mathbb{N}^+$.

Output: Approximation $\tilde{\tau}_k$ to moment $\frac{1}{n} \text{tr}(\bar{T}_k(A))$ for all $k \in 1, \dots, N$.

- 1: Draw $g^{(1)}, \dots, g^{(\ell)} \sim \text{Uniform}(\{-1, 1\}^n)$.
 - 2: For $k = 1, \dots, N$, $\tilde{\tau}_k \leftarrow \frac{\sqrt{2/\pi}}{\ell n} \sum_{i=1}^{\ell} (g^{(i)})^\top T_k(A) g^{(i)}$. $\triangleright$ Computed using recurrence in (8)
 - 3: Return $\tilde{\tau}_1, \dots, \tilde{\tau}_N$.
-

Lemma 5.2. *If Algorithm 3 is run with $\ell = \max\left(1, C \cdot \log^2(N/\delta)/(n\Delta^2)\right)$, where C is a fixed positive constant, then with probability $1 - \delta$ the approximate moments returned satisfy $|\tilde{\tau}_k - \frac{1}{n} \text{tr}(\bar{T}_k(A))| \leq \Delta$ for all $k = 1, \dots, N$.*

Proof. Fix $k \in \{1, \dots, N\}$. Note that $\frac{1}{n} \text{tr}(\bar{T}_k(A)) = \frac{\sqrt{2/\pi}}{n} \text{tr}(T_k(A))$. Let C be the constant from Lemma 5.1. If $\ell = \max\left(1, C^2 \cdot \log^2(N/\delta)/(n\Delta^2)\right)$, then by that lemma we have that with probability at least $1 - \delta/N$:

$$\left| \tilde{\tau}_k - \frac{\sqrt{2/\pi}}{n} \text{tr}(T_k(A)) \right| \leq \frac{1}{n} \frac{C \log(N/\delta)}{\sqrt{\ell}} \|T_k(A)\|_F \leq \frac{C \sqrt{2/\pi}}{\sqrt{n}} \sqrt{\frac{\log(N/\delta)}{\ell}} \leq \Delta.$$

The second to last inequality follows from the fact that $\|T_k(A)\|_2 \leq 1$ and thus $\|T_k(A)\|_F \leq \sqrt{n}$. Applying a union bound over all $k \in 1, \dots, N$ gives the claim. $\square$

Theorem 1.1 immediately follows as a corollary of Lemma 5.2 and Lemma 4.1.

Proof of Theorem 1.1. We implement Algorithm 2 with Algorithm 3 used as a subroutine to approximate the Chebyshev polynomial moments, which requires setting $\Delta = \frac{1}{N^2}$. By Lemma 5.2, we conclude that we need to set $\ell = \max\left(1, CN^2 \log^2(N/\delta)/n\right)$. Then, by Lemma 4.1, setting $N = O(1/\epsilon)$ ensures that Algorithm 2 returns a distribution q which is ϵ close to A 's spectral density s in Wasserstein distance. $\square$

5.2 Approximate Matrix-Vector Multiplications

Algorithm 3 assumes access to an oracle for computing exact matrix-vector multiplies with A . In this section, we show that the method continues to work well even when each term in Hutchinson's estimator, $g^\top T_k(A)g$, is computed using an approximate matrix-vector multiplication oracle for A (see Definition 1.2). As discussed in Section 1.1, the robustness of the estimator allows the kernel polynomial method to be applied in many settings where A can only be accessed implicitly. It also forms the basis of our sublinear time algorithm for computing the spectral density of a normalized graph adjacency or Laplacian matrix, which are presented in the Section 6.

To show that approximate matrix-vector multiplications suffice, we leverage well understood stability properties of the three-term forward recurrence for Chebyshev polynomials of the first kind [Cle55, MMS18]. These properties allow us to analyze the cumulative error when $T_k(A)g$ is computed via this recurrence. Specifically, we analyze the following algorithm:

Algorithm 4 Hutchinson Moment Estimator w/ Approximate Multiplications

Input: Diagonalizable $A \in \mathbb{R}^{n \times n}$ with $\|A\|_2 \leq 1$, degree $N \in 4\mathbb{N}^+$, number of repetitions $\ell \in \mathbb{N}^+$, ϵ_{MV} -approximate matrix vector multiplication oracle AMV for A (see Definition 1.2).

Output: Approximation $\tilde{\tau}_k$ to moment $\frac{1}{n} \text{tr}(\bar{T}_k(A))$ for all $k \in 1, \dots, N$.

- 1: **for** $i = 1, \dots, \ell$ iterations **do**
 - 2: Draw $g \sim \text{Uniform}(\{-1, 1\}^n)$.
 - 3: $\tilde{v}_0 \leftarrow g, \tilde{v}_1 \leftarrow \text{AMV}(A, g, \epsilon_{\text{MV}})$.
 - 4: $\tilde{\tau}_{1,i} \leftarrow g^T \tilde{v}_1$
 - 5: **for** $k = 2$ to N **do**
 - 6: $\tilde{v}_k \leftarrow 2 \cdot \text{AMV}(A, \tilde{v}_{k-1}, \epsilon_{\text{MV}}) - \tilde{v}_{k-2}$.
 - 7: $\tilde{\tau}_{k,i} \leftarrow g^T \tilde{v}_k$
 - 8: **For** $k = 1, \dots, N, \tilde{\tau}_k \leftarrow \frac{1}{\ell} \sum_{i=1}^{\ell} \tilde{\tau}_{k,i}$.
 - 9: **Return** $\tilde{\tau}_1, \dots, \tilde{\tau}_N$.
-

Algorithm 4 assumes access to an approximate matrix-vector multiplication oracle for A with error ϵ_{MV} (recall Definition 1.2). Since $\|A\|_2 \leq 1$, for any vector y , we have that:

$$\|\text{AMV}(A, y, \epsilon_{\text{MV}}) - Ay\|_2 \leq \epsilon_{\text{MV}} \|y\|_2. \quad (9)$$

The algorithm uses this oracle to apply the recurrence from (8), approximately computing each $T_k(A)g$ for $k = 1, \dots, N$, which in turn allows us to approximately compute $g^T T_k(A)g$. Note that when $\epsilon_{\text{MV}} = 0$, Algorithm 4 is exactly equivalent to Algorithm 3.

Notation. Analyzing this approach requires accounting for error accumulates across iterations. To do so, we introduce some basic notation. Let v_k denote the true value of $T_k(A)g$, and let $\tilde{v}_k$ denote our computed approximation. We initialize the recurrence with $\tilde{v}_{-1} = \vec{0}$ and $\tilde{v}_0 = v_0 = g$. For $k = 0, \dots, N-1$, let $w_k = \text{AMV}(A, \tilde{v}_k, \epsilon_{\text{MV}})$ and note that $\|w_k - A\tilde{v}_k\|_2 \leq \epsilon_{\text{MV}} \|\tilde{v}_k\|_2$. In iteration k of the recurrence, we compute $\tilde{v}_{k+1}$ by applying the recurrence:

$$\tilde{v}_{k+1} := 2w_k - \tilde{v}_{k-1}.$$

For each $i \in 0, \dots, N$ we denote:

- $\delta_k := v_k - \tilde{v}_k$, with $\delta_0 = \vec{0}$. This is the *accumulated error* up to iteration k .
- $\xi_{k+1} := A\tilde{v}_k - w_k$, with $\xi_0 = 0$. $2\xi_{k+1}$ is the *new error* introduced in iteration k due to approximate matrix-vector multiplication.

As in Clenshaw's classic work [Cle55], it can be shown that δ_k itself evolves according to a *simple recurrence*, which ultimately lets us show that it can be expressed as a summation involving Chebyshev polynomials of the *second* kind, which are easily bounded. Specifically, we have:

Fact 5.3. $\delta_1 = \xi_1$ and for $2 \leq k \leq N$, $\delta_k = 2A\delta_{k-1} - \delta_{k-2} + 2\xi_k$.

Proof. The claim for δ_1 is direct since $v_0 = \tilde{v}_0$: we have $\delta_1 = v_1 - \tilde{v}_1 = Av_0 - w_0$. For $2 \leq k \leq N$, we prove the claim by writing the difference $\delta_k = v_k - \tilde{v}_k = v_k - 2(A\tilde{v}_{k-1} + \xi_k) + \tilde{v}_{k-2}$. We can then replace $v_k = 2Av_{k-1} - v_{k-2}$ and substitute in $(v_{k-1} - \tilde{v}_{k-1}) = \delta_{k-1}$ and $(v_{k-2} - \tilde{v}_{k-2}) = \delta_{k-2}$. $\square$

The Chebyshev polynomials of the second kind are defined via the following recurrence:

Definition 5.4 (Chebyshev Polynomials of the Second Kind). For $k \in \mathbb{N}^{\geq 0}$ the k -th Chebyshev polynomial of the second kind $U_k(x)$ is given by

$$\begin{aligned} U_0(x) &= 1 & U_1(x) &= 2x \\ U_k(x) &= 2x \cdot U_{k-1}(x) - U_{k-2}(x) & \text{for } k \geq 2. \end{aligned}$$

We also define $U_{-1}(x) = 0$, which is consistent with the recurrence.

Using these polynomials, we can characterize the accumulated error δ_k in terms of the error introduced in each of the prior iterations.

Lemma 5.5. For $k = 1, \dots, N$, we have

$$\delta_k = U_{k-1}(A)\xi_1 + 2 \sum_{i=2}^k U_{k-i}(A)\xi_i. \quad (10)$$

Proof. We prove the lemma by induction on $j \leq k$. For $j = 0$, the lemma is trivial since $\delta_0 = 0$ by definition and $U_{-1}(A) = 0$. For $j = 1$, $\delta_1 = \xi_1 = U_0(A)\xi_1$. By Fact 5.3, for $2 \leq j < k$, we have:

$$\delta_j = 2\xi_j + \underbrace{2A\delta_{j-1} - \delta_{j-2}}_{z_1}. \quad (11)$$

We can apply the inductive hypothesis on z_1 and recombine terms using Definition 5.4 to get:

$$\begin{aligned} z_1 &= 2A \cdot \left(U_{j-2}(A)\xi_1 + 2 \sum_{i=2}^{j-1} U_{j-1-i}(A)\xi_i \right) - U_{j-3}(A)\xi_1 - 2 \sum_{i=2}^{j-2} U_{j-2-i}(A)\xi_i \\ &= U_{j-1}(A)\xi_1 + U_1(A) \cdot 2\xi_{j-1} + \sum_{i=2}^{j-2} (2AU_{j-1-i}(A) - U_{j-2-i}(A)) \cdot 2\xi_i \\ &= U_{j-1}(A)\xi_1 + \sum_{i=2}^{j-1} U_{j-i}(A) \cdot 2\xi_i \end{aligned}$$

Noting that plugging into (11) and noting that $2\xi_j = 2U_0(A)\xi_j$ completes the proof. $\square$

Our goal is to use Lemma 5.5 to establish that δ_k is small because each ξ_i is small. It is well known that the Chebyshev polynomials of the second kind satisfy the following bounds for any $k \in \mathbb{N}$:

$$|U_k(x)| \leq k + 1 \quad \text{for} \quad x \in [-1, 1]. \quad (12)$$

This is the upper bound we need to proceed. Specifically, we will show that each estimator using Algorithm 4, $g^\top \tilde{v}_k$, well approximates Hutchinson's estimator $g^\top T_k(A)g = g^\top v_k$.

Claim 5.6. For quantities $v_k, \tilde{v}_k$ and $0 \leq \epsilon_{MV} \leq 1/2k^2$, we have

$$\left| g^\top T_k(A)g - g^\top \tilde{v}_k \right| \leq 2\epsilon_{MV} \cdot (k+1)^2 \|g\|_2^2.$$

Proof. By the definition of δ_k , we have $|g^\top T_k(A)g - g^\top \tilde{v}_k| = |g^\top \delta_k|$. By Cauchy-Schwarz we can bound $|g^\top \delta_k| \leq \|g\|_2 \|\delta_k\|_2$. We are left to bound $\|\delta_k\|_2$. Applying Lemma 5.5 and triangle inequality, we have

$$\|\delta_k\|_2 \leq \|U_{k-1}(A)\|_2 \|\xi_1\|_2 + \sum_{i=2}^k 2\|U_{k-i}(A)\|_2 \|\xi_i\|_2$$

Then applying (12) and the fact that $\|A\|_2 \leq 1$, we have $\|U_{k-i}(A)\|_2 \leq (k-i+1)$. Hence,

$$\|\delta_k\|_2 \leq k\|\xi_1\|_2 + \sum_{i=2}^k 2(k-i+1)\|\xi_i\|_2 \leq \sum_{i=1}^k 2(k-i+1)\|\xi_i\|_2.$$

Using that $\xi_i \leq \epsilon_{\text{MV}} \|\tilde{v}_{i-1}\|_2$, and that $\|T_i(A)\|_2 \leq 1$ for all i and thus $\|v_i\|_2 \leq \|g\|_2$, we have:

$$\begin{aligned} \|\delta_k\|_2 &\leq \sum_{i=1}^k 2(k-i+1) \epsilon_{\text{MV}} \|\tilde{v}_{i-1}\|_2 \leq 2\epsilon_{\text{MV}} \sum_{i=1}^k (k-i+1) (\|v_{i-1}\|_2 + \|\delta_{i-1}\|_2) \\ &\leq \epsilon_{\text{MV}} k(k+1) \left(\|g\|_2 + \max_{i \leq k} \|\delta_i\|_2 \right). \end{aligned}$$

Inducting on δ_j for $j \leq k$ gives us $\|\delta_k\|_2 \leq 2\epsilon_{\text{MV}}(k+1)^2 \|g\|_2$, which completes the proof. $\square$

Lemma 5.7. *If Algorithm 4 is run with $\ell = \max\left(1, C \cdot \log^2(N/\delta)/(n\Delta^2)\right)$ and $\epsilon_{\text{MV}} = \Delta/4N^2$, where C is a fixed positive constant, then with probability $1 - \delta$ the approximate moments returned satisfy $|\tilde{\tau}_k - \frac{1}{n} \text{tr}(\bar{T}_k(A))| \leq \Delta$ for all $k = 1, \dots, N$.*

Proof. Fix $k \in \{1, \dots, N\}$. Let $g^{(1)}, \dots, g^{(\ell)}$ be the random vectors drawn in the outer for-loop of Algorithm 4. Let $\{\tilde{v}_k^{(i)}\}_{i \in [\ell]}$ be the ℓ vectors computed by the inner for-loop and let $\{\delta_k^{(i)} := \tilde{v}_k^{(i)} - T_k(A)g^{(i)}\}_{i \in [\ell]}$ be the ℓ error vectors. Recalling that $\frac{1}{n} \text{tr}(\bar{T}_k(A)) = \frac{\sqrt{2/\pi}}{n} \text{tr}(T_k(A))$, we have:

$$\left| \tilde{\tau}_k - \frac{\sqrt{2/\pi}}{n} \text{tr}(T_k(A)) \right| \leq \frac{\sqrt{2/\pi}}{n\ell} \sum_{i=1}^{\ell} |(g^{(i)})^\top \delta_k^{(i)}| + \left| \frac{\sqrt{2/\pi}}{n\ell} \sum_{i=1}^{\ell} (g^{(i)})^\top T_k(A)g^{(i)} - \frac{1}{n} \text{tr}(T_k(A)) \right|$$

Applying Claim 5.6 and Lemma 5.1, with probability at least $1 - \delta/N$, we thus have

$$|\tilde{\tau}_k - \frac{1}{n} \text{tr}(\bar{T}_k(A))| \leq 2(k+1)^2 \epsilon_{\text{MV}} \cdot \frac{\sqrt{2/\pi}}{n\ell} \sum_{i=1}^{\ell} \|g^{(i)}\|_2^2 + \Delta/2 \leq \Delta/2 + \Delta/2.$$

The last inequality follows from the fact that $\|g^{(i)}\|_2^2 = n$ for all $i \in [\ell]$, and the choice of $\epsilon_{\text{MV}} = \Delta/4N^2$. Applying a union bound over all $k = 1, \dots, N$ gives the claim. $\square$

Theorem 1.3 immediately follows.

Proof of Theorem 1.3. We implement Algorithm 2 with Algorithm 4 used as a subroutine to approximate the Chebyshev polynomial moments, which requires setting $\Delta = \frac{1}{N^2}$. By Lemma 5.7, we conclude that we need to set $\ell = \max\left(1, CN^2 \log^2(N/\delta)/n\right)$ and $\epsilon_{\text{MV}} = 1/4N^4$. Then, by Lemma 4.1, setting $N = O(1/\epsilon)$ ensures that Algorithm 2 returns a distribution q which is ϵ close to A 's spectral density s in Wasserstein distance. $\square$

6 Sublinear Time Methods for Graphs

An interesting application of Theorem 1.3 is that it allows for the kernel polynomial method of Algorithm 2 to be combined with any randomized algorithm for approximating matrix-vector multiplications with A . In this section we show that, for a common class of matrices derived from undirected graphs, such an algorithm can actually be implemented in *sublinear time*, leading to a sublinear time SDE algorithm for computing graph spectra from these matrices.

Let $A \in \mathbb{R}^{n \times n}$ be the normalized adjacency matrix for an unweighted, n -vertex graph $G = (V, E)$ with spectral density s . I.e., A is the standard adjacency matrix multiplied on the left and right by $D^{-1/2}$, where D is an $n \times n$ diagonal matrix containing the degree of each node in V . We assume a model where we are given access to these degrees, $[d_1, \dots, d_n]$.

Under this assumption we show that, for any $\epsilon_{\text{MV}} \in (0, 1)$ and failure probability $\delta \in (0, 1)$, an ϵ_{MV} -approximate matrix-vector multiplication oracle for A can be implemented in $O(n \epsilon_{\text{MV}}^{-2} \log^2(1/\delta))$ time. Via Theorem 1.3, this immediately yields an algorithm for computing an SDE that is ϵ close in Wasserstein-1 distance to A 's spectral density in roughly $O(n/\epsilon^9)$ time for sufficiently large n , and at most $\tilde{O}(n/\epsilon^{13})$ time. Our main result is stated as Theorem 1.4 in Section 1.1.

The same algorithm can be used to approximate the spectral density of the normalized Laplacian of G by a simple shift and scaling. Specifically, A can be obtained from the normalized Laplacian L via $A = I - L$, and the spectral density of L , $s_L(x)$ satisfies $s_L(1 - x) = s_A(x)$, where s_A is the spectral density of A . So if we obtain an ϵ -approximate SDE q for A by Theorem 1.4, then the function p satisfying $p(1 - x) = q(x)$ is an ϵ -approximate SDE for s_L . We thus have:

Corollary 6.1. *Given the normalized adjacency matrix of G , there exists an algorithm that takes $O(n \text{poly}(\frac{\log(1/\delta)}{\epsilon}))$ time and outputs a density function q that is ϵ close to the spectral density of the normalized Laplacian of G with probability at least $1 - \delta$.*

6.1 Approximate Matrix-Vector Multiplication for Adjacency Matrices

We implement an ϵ_{MV} -approximate matrix-vector multiplication oracle for A using a randomized column-sampling based algorithm of [DKM06]. Throughout this section we denote the i -th column of A by A^i . Given the matrix A , vector $y \in \mathbb{R}^n$ and a sampling budget $t \in \mathbb{N}$, the algorithm samples and rescales t columns of A independently and with replacement. To approximate Ay , the algorithm outputs the multiplication of these sampled columns with the appropriate elements of y .

Algorithm 5 AMV Multiplication Oracle for Normalized Adjacency Matrices

Input: Normalized adjacency matrix $A \in \mathbb{R}^{n \times n}$, degrees $[d_1, \dots, d_n]$, vector y , and parameter $t \in \mathbb{N}$.

Output: A vector $z \in \mathbb{R}^n$ that approximates Ay .

- 1: Initialize multiset $\mathcal{S} \leftarrow \emptyset$
 - 2: **for** t iterations **do**
 - 3: Add random index $i \in 1, \dots, n$ to $\mathcal{S}$, where i is sampled with prob. $p_i = \frac{\|A^i\|_2^2}{\|A\|_F^2} = \frac{1/d_i}{\sum_{j=1}^n 1/d_j}$
 - 4: **return** $z \leftarrow \sum_{i \in \mathcal{S}} \frac{1}{tp_i} \cdot y_i A^i$
-

We state a lemma by [DKM06], which bounds the variance Algorithm 5's error.

Lemma 6.2 (Lemma 4, [DKM06] – rephrased for our setting). *Let $A \in \mathbb{R}^{n \times n}$ be a matrix, $y \in \mathbb{R}^n$ be a vector and $t \in \mathbb{N}^+$ be a fixed constant. Let $z \in \mathbb{R}^n$ be the output of Algorithm 5 after sampling t indices with probabilities $[p_1, \dots, p_n]$, then*

$$\mathbf{E}[\|Ay - z\|_2^2] = \sum_{i=1}^n \frac{y_i^2 \|A^i\|_2^2}{tp_i} - \frac{1}{t} \|Ay\|_2^2.$$

Using this lemma, we show that there is an ϵ_{MV} -approximate matrix-vector oracle for A based on Algorithm 5 with success probability at least $1 - \delta$ that runs in $O(n \epsilon_{MV}^{-2} \log^2(\frac{1}{\delta}))$ time.

Proposition 6.3. *Let $A \in \mathbb{R}^{n \times n}$ be the normalized Laplacian of an n -vertex graph $G = (V, E)$ and let $\epsilon_{MV}, \delta \in (0, 1)$ be fixed constants. There is an algorithm that given a vector $y \in \mathbb{R}^n$ and access to the degrees of the vertices of G takes $O(n \epsilon_{MV}^{-2} \log^2(\frac{1}{\delta}))$ time and outputs a vector $z \in \mathbb{R}^n$ such that $\|z - Ay\|_2 \leq \epsilon_{MV} \|y\|_2$ with probability at least $1 - \delta$.*

Proof. Notice that $\|A^i\|_2^2 = \frac{1}{d_i}$ since A is a normalized adjacency matrix. Fix $t = 48 \epsilon_{MV}^{-2} \sum_{i=1}^n \frac{1}{d_i} = O(\epsilon_{MV}^{-2} \|A\|_F^2)$. Then, by Lemma 6.2 and Chebyshev's inequality, we have that when Algorithm 5 is called on A with parameter t ,

$$\Pr[\|Ay - z\|_2 > \frac{\epsilon_{MV}}{4} \|y\|_2] \leq \frac{16 \sum_{i=1}^n y_i^2 \|A^i\|_2^2}{t \epsilon_{MV}^2 \|y\|_2^2} \leq \frac{1}{4}. \quad (13)$$

In order to get $1 - \delta$ success probability, we repeat the above process $r = c \log(\frac{1}{\delta})$ times for an independent constant c to be fixed later. Let $z_1, \dots, z_r \in \mathbb{R}^n$ be the output of running Algorithm 5 r times with parameter t . We can then pick the first z_i such that $\|z_i - z_j\|_2 \leq \frac{\epsilon_{MV}}{2} \|y\|_2$ for at least $r/2 + 1$ vectors z_j from $z_1, \dots, z_r$.

To see why this works, let us fix a large enough constant c so we can use a Chernoff bound to claim that with probability at least $1 - \delta$ at least $r/2 + 1$ vectors z_j from $z_1, \dots, z_r$ have that $\|z_j - Ay\|_2 \leq \frac{\epsilon_{MV}}{4} \|y\|_2$. By a triangle inequality we have that for all such z_j and z_k ,

$$\|z_j - z_k\|_2 \leq \|z_j - Ay\|_2 + \|z_k - Ay\|_2 \leq \frac{\epsilon_{MV}}{2} \|y\|_2.$$

Thus, the z_i we picked must satisfy that $\|z_i - Ay\|_2 \leq \frac{3\epsilon_{MV}}{4} \|y\|_2$ by the triangle inequality. Since we sample columns independently and without replacement, the sampling of the indices can be done in $O(n + rt) < O(rn)$ time.

Next, we bound, in expectation, the total number of non-zero entries in the sampled columns $\{A^i\}_{i \in \mathcal{S}}$ of one call to Algorithm 5 with parameter $t = 48 \epsilon_{MV}^{-2} \sum_{i=1}^n \frac{1}{d_i}$. This will dictate the runtime of the algorithm, as the time to compute z is linear in this number. Importantly, we need to take advantage of the fact that the algorithm samples sparse columns with higher probability than dense columns – thus, while many columns may be selected, their total sparsity will be low.

$$\mathbf{E} \left[\sum_{i \in \mathcal{S}} \text{nnz}(A^i) \right] = t \sum_{i=1}^n d_i \frac{\|A^i\|_2^2}{\|A\|_F^2} = \frac{tn}{\|A\|_F^2} = 48n \epsilon_{MV}^{-2}.$$

The final equality follows from the fact that $t = 48 \epsilon_{MV}^{-2} \|A\|_F^2$.

We can get $1 - \delta$ success probability for the number of non-zeros in the sampled columns by having a large enough budget $b = 96 \epsilon_{\text{MV}}^{-2}$, repeatedly running Algorithm 5 until the number of non-zero entries in the sampled columns is within the budget and throwing out a run of the algorithm if it exceeds the budget. By Markov’s inequality, with probability $1/2$, the number of non-zero entries will not exceed b . So with probability at least $1 - \delta$ we will have to re-run the algorithm only $O(\log(\frac{1}{\delta}))$ times. We can take a union bound on the failure condition in (13) and the number of non-zero entries being within the budget b .

Overall, our approach need to make $O(\log^2(\frac{1}{\delta}))$ calls to Algorithm 5. The r^2 pairwise comparisons can be done in nr^2 time and the running time of computing the output vector in Algorithm 5 is proportional to the number of non-zeros in the sampled columns, i.e. $O(n \epsilon_{\text{MV}}^{-2})$. $\square$

Proof of Theorem 1.4. The accuracy and running time claim follows from combining the ϵ_{MV} -approximate vector multiplication oracle described in Proposition 6.3 with Algorithm 2, which is analyzed in Theorem 1.3. $\square$

As discussed in the introduction, Cohen et al. [CKSV18] prove a result which matches the guarantee of Theorem 1.4, but with runtime of $2^{O(1/\epsilon)}$ – i.e., with *no dependence* on n . In comparison, our result depends linearly on n , but only polynomially on $1/\epsilon$. In either case, the result is quite surprising, as the runtime is *sublinear* in the input size: A could have up to $O(n^2)$ non-zero entries.

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A Approximate Eigenvalues from Spectral Density Estimate

Algorithm 1 and Algorithm 2 in the previous sections output a closed form representation of a distribution q which is close in Wasserstein-1 distance to s . In particular, the distribution output is continuous. Alternatively, we describe an algorithm from [CKSV18] (Algorithm 6) that recovers a list of n eigenvalues $\tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$ such that $\|\Lambda - \tilde{\Lambda}\|_1 \leq 2n\epsilon$, which implies that the discrete distribution associated with $\tilde{\Lambda}$ is 2ϵ close to s in Wasserstein-1 distance. Formally:

Theorem A.1. *Let s be a spectral density and let q be a density on $[-1, 1]$ such $W_1(s, q) \leq \epsilon$ for $\epsilon \in (0, 1)$. As long as q can be integrated over any subinterval of $[-1, 1]$ (e.g., has a closed form antiderivative), there is an algorithm (Algorithm 6) that computes $1/\epsilon$ such integrals and in $O(n + 1/\epsilon)$ additional time outputs a list of n values $\tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$ such that $\|\Lambda - \tilde{\Lambda}\|_1 \leq 2n\epsilon$.*

At a high-level, the algorithm computes a grid with spacing ϵ for the interval $[-1, 1]$, “snaps” the mass of the continuous density onto the nearest point in the net, and then readjusts the resulting point masses to a distribution where every point mass is divisible by $1/n$ (and can therefore be represented by a certain number of eigenvalues). We describe the method in detail in Algorithm 6.

The method requires computing the mass $\int_a^b q(x)dx$ where $-1 \leq a < b \leq 1$. For Algorithms 1 and 2, q is written as $q = w \cdot p$ where p is a degree N polynomial written as a sum of the first $N + 1$ Chebyshev polynomials. So to compute this integral, we need to compute the integral $\int_a^b T_k(x)w(x)dx$ for any $k \in 1, \dots, N$ gives a method for computing $\int_a^b q(x)dx$. We give a closed form expression for this integral below:

Fact A.2. *For $k \in \mathbb{N}^{>0}$ and $-1 \leq a < b \leq 1$ we have that*

$$\int_a^b \frac{T_k(x)}{\sqrt{1-x^2}} dx = \frac{-\cos(ku)}{k} \Big|_{\sin^{-1}a}^{\sin^{-1}b}$$

Algorithm 6 Approximate Eigenvalues from Spectral Density

Input: Spectral density $q : [-1, 1] \rightarrow \mathbb{R}^+$, integer n .

Output: Vector $\tilde{\Lambda} = [\tilde{\lambda}_1, \dots, \tilde{\lambda}_n]$.

- 1: compute $\vec{v} = (v_{-1+\epsilon}, v_{-1+2\epsilon}, \dots, v_0, v_\epsilon, \dots, v_1)$ such that $v_t = \int_{t-\epsilon}^t q(x)dx$
- 2: define $f_v : [0, 1] \rightarrow [-1, 1]$ to be a non-decreasing step-function with the property that for $X \sim \text{Uniform}([0, 1])$, $f_v(X) = -1 + t\epsilon$ with probability $v_{-1+t\epsilon}$ for $t \in [1, \dots, \frac{2}{\epsilon}]$.
- 3: Set

$$\tilde{\lambda}_i = \mathbf{E} \left[f_v(X) \mid X \in \left[\frac{i-1}{n}, \frac{i}{n} \right] \right]$$

To prove Theorem A.1, we first state a proposition from [CKSV18], which we prove for completeness below. It shows that, for any equally weighted point-mass distribution, the Wasserstein-1 distance to the point-mass distribution $\tilde{\Lambda}$ is at most the distance to the distribution $\vec{v}$.

Proposition A.3 (Lemma 8, [CKSV18]). *Let p be a distribution of n equally weighted point-masses on points $p_1, \dots, p_n$ with $p_i \leq p_{i+1}$. Let v be the distribution defined by Algorithm 6 on the grid $(-1 + \epsilon, \dots, 0, \epsilon, \dots, 1)$ with probabilities $\vec{v}$. Let $\tilde{\Lambda}$ be the output of Algorithm 6 and let l denote the distribution corresponding to having n equally weighted point masses on the points in $\tilde{\Lambda}$. We then have that $W_1(p, l) \leq W_1(p, v)$.*

Proof of Theorem A.1. Consider the output $\tilde{\Lambda}$ of Algorithm 6 with input q and n . Notice that $W_1(v, q) \leq \epsilon$ by the definition of v and the earthmover's definition of the Wasserstein distance. Hence, by triangle inequality, we have that $W_1(v, s) \leq 2\epsilon$. Define l to be the distribution corresponding to having n equally weighted point-masses on the points in $\tilde{\Lambda}$. By Proposition A.3 we know that $\|\Lambda - \tilde{\Lambda}\|_1 = W_1(s, l) \leq W_1(v, s) \leq 2\epsilon$.

By Fact A.2 we can compute the distribution v exactly by performing the integration in closed form. Each integral takes $O(1/\epsilon)$ time for a total of $O(1/\epsilon^2)$ time. The final operation, Line 3, takes $O(n)$ time. $\square$

Proof of Proposition A.3. By the earthmover's distance definition of the Wasserstein-1 distance, the minimal cost mass-moving scheme that yields distribution p from v corresponds to moving the $\frac{1}{n}$ probability mass from the conditional distribution $f_v(X)|X \in [\frac{i-1}{n}, \frac{i}{n}]$ to location p_i . Notice that $W_1(p, \tilde{\Lambda}) = \frac{1}{n} \sum_{i=1}^n |p_i - \tilde{\lambda}_i|$. It is sufficient then to analyze $|p_i - \tilde{\lambda}_i|$ independently for each i .

$$\begin{aligned} W_1(p, v) &= \frac{1}{n} \sum_{i=1}^n \sum_{t=1}^{2/\epsilon} |(-1 + t\epsilon) - p_i| \cdot \Pr \left[f_v(X) = (-1 + t\epsilon) \mid X \in \left[\frac{i-1}{n}, \frac{i}{n} \right] \right] \\ &\geq \frac{1}{n} \sum_{i=1}^n \left| p_i - \sum_{t=1}^{2/\epsilon} (-1 + t\epsilon) \cdot \Pr \left[f_v(X) = (-1 + t\epsilon) \mid X \in \left[\frac{i-1}{n}, \frac{i}{n} \right] \right] \right| \\ &= \frac{1}{n} \sum_{i=1}^n |p_i - \tilde{\lambda}_i| = W_1(p, l). \end{aligned}$$

$\square$

B Positive Polynomial Approximation

In this section, we introduce Jackson's powerful result from 1912 on the uniform approximation of Lipschitz continuous periodic functions by low-degree trigonometric polynomials [Jac12, Jac30]. This result will directly translate to the result for algebraic polynomials needed to analyze the kernel polynomial method. We start with basic definitions and preliminaries below.

B.1 Fourier Series Preliminaries

Definition B.1 (Fourier Series). A function f with period 2π that is integrable on the length of that period can be written via the Fourier series:

$$f(x) = \frac{\alpha_0}{2} + \sum_{k=1}^{\infty} \alpha_k \cos(kx) + \beta_k \sin(kx)$$

where

$$\alpha_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) dx \quad \beta_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx) dx.$$

Equivalently we can write f in *exponential form* as:

$$f(x) = \sum_{k=-\infty}^{\infty} \hat{f}_k e^{ikx}$$

where $i = \sqrt{-1}$, $\hat{f}_0 = \alpha_0/2$, $\hat{f}_k = \hat{f}_{|k|}^*$ for $k < 0$, and for $k > 0$,

$$\hat{f}_k = \frac{1}{2}(\alpha_k - i\beta_k).$$

If the Fourier series of a periodic function f has $\hat{f}_k = 0$ for $k > N$ (equivalently, $\alpha_k = \beta_k = 0$ for $k > N$), we say that f is a degree N trigonometric polynomial.

In working with Fourier series, we require the two standard convolution theorems:

Claim B.2 (First Convolution Theorem). *Let f, g be integrable 2π -periodic functions with exponential form Fourier series coefficients $[\hat{f}_k]_{k=-\infty}^{\infty}$ and $[\hat{g}_k]_{k=-\infty}^{\infty}$, respectively. Let h be their convolution:*

$$h(x) = [f * g](x) = \int_{-\pi}^{\pi} f(u)g(x-u)du.$$

The exponential form Fourier series coefficients of h , $[\hat{h}_k]_{k=-\infty}^{\infty}$, satisfy:

$$\hat{h}_k = 2\pi \cdot \hat{f}_k \hat{g}_k$$

Claim B.3 (Second Convolution Theorem). *Let f, g be integrable 2π -periodic functions with exponential form Fourier series coefficients $[\hat{f}_k]_{k=-\infty}^{\infty}$ and $[\hat{g}_k]_{k=-\infty}^{\infty}$, respectively. Let h be their product:*

$$h(x) = f(x) \cdot g(x).$$

The exponential form Fourier series coefficients of h , $[\hat{h}_k]_{k=-\infty}^{\infty}$, satisfy:

$$\hat{h}_k = \sum_{j=-\infty}^{\infty} \hat{f}_j \cdot \hat{g}_{k-j}$$

In other words, the Fourier coefficients of h are the discrete convolution of those of f and g .

B.2 Jackson's Theorem for Trigonometric Polynomials

We seek a low-degree trigonometric polynomial $\tilde{f}$ that is a good *uniform* approximation to any sufficiently smooth periodic function f . I.e., we want $\|f - \tilde{f}\|_{\infty} < \epsilon$ where $\|z\|_{\infty}$ denotes $\|z\|_{\infty} = \max_x z(x)$. A natural choice for $\tilde{f}$ is the truncated Fourier series $\sum_{k=-N}^N c_k e^{ikx}$, but this does not lead to good uniform approximation in general. Instead, Jackson showed that better accuracy can be obtained with a Fourier series with *damped coefficients*, which is equivalent to the *convolution* of f with an appropriately chosen “bump” function (aka kernel), defined below:

Definition B.4 (Jackson Kernel). For any positive integer m , let b be the $2m - 2$ degree trigonometric polynomial:

$$b = \left(\frac{\sin(mx/2)}{\sin(x/2)} \right)^4 = \sum_{k=-2m+2}^{2m-2} \hat{b}_k e^{ikx},$$

which has exponential form coefficients $\hat{b}_{-2m+2}, \dots, \hat{b}_0, \dots, \hat{b}_{2m-2}$ equal to:

$$\hat{b}_{-k} = \hat{b}_k = \sum_{j=-m}^{m-k} (m - |j|) \cdot (m - |j + k|) \quad \text{for } k = 0, \dots, 2m - 2. \quad (14)$$

When m is odd it is easy to see that b is a degree $2m - 2$ trigonometric polynomial. Specifically, for odd m we have the well known Fourier series of the periodic sinc function $s(x) = \frac{\sin(mx/2)}{\sin(x/2)} = \sum_{k=-(m-1)/2}^{(m-1)/2} e^{ikx}$. We then apply the convolution theorem (Claim B.3) to $s(x) \cdot s(x)$. to see that $s^2(x) = \left(\frac{\sin(mx/2)}{\sin(x/2)}\right)^2$ is an $m-1$ degree trigonometric polynomial with coefficients $c_{-k} = c_k = m-k$. Applying it again to $s^2(x) \cdot s^2(x)$ yields (14). For a derivation of (14) when m is even, we refer the reader to [Jac30] or [Lor66].

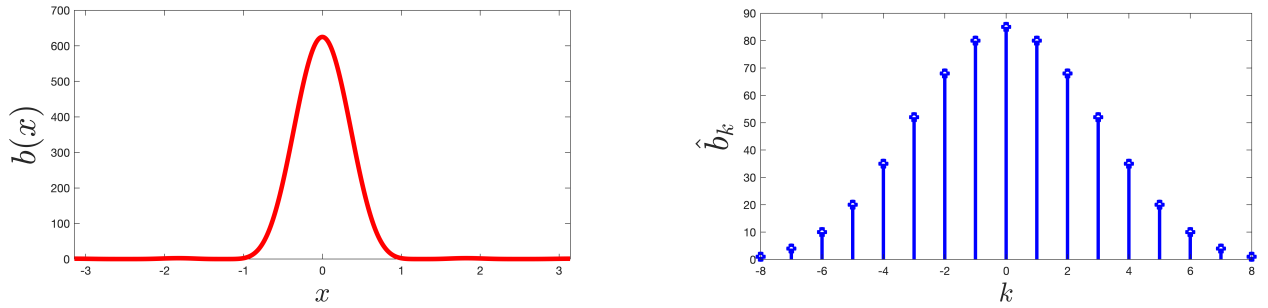


Figure 3: Jackson's bump function $b(x)$ for $m = 5$, alongside its Fourier series coefficients.

Jackson's main result is as follows. We include a short proof for completeness.

Theorem B.5 (Jackson [Jac12], see also [Jac30]). *Let f be a 2π -periodic, Lipschitz continuous function with Lipschitz constant λ . I.e., $|f(x) - f(y)| \leq \lambda|x - y|$ for all x, y . For integer m , let b be the bump function from Definition 14, with k^{th} Fourier coefficients $\hat{b}_k$. The function $\tilde{f}(x) = \frac{1}{2\pi\hat{b}_0} \int_{-\pi}^{\pi} b(u)f(x - u)du$ satisfies:*

$$\|\tilde{f} - f\|_{\infty} \leq 9 \frac{\lambda}{m}.$$

$\tilde{f}$ is a $2m - 2$ degree trigonometric polynomial, and by the convolution theorem, its exponential form Fourier series coefficients are given by $\hat{\tilde{f}}_k = \frac{\hat{b}_k}{\hat{b}_0} \cdot \hat{f}_k$ for $k = -2m + 2, \dots, 2m - 2$.

Remark. The function $\tilde{f}$ takes the form of a *damped* truncation of f 's Fourier series: $\frac{\hat{b}_0}{\hat{b}_0} = 1$ and $\frac{\hat{b}_k}{\hat{b}_0}$ falls off towards zero as $k \rightarrow 2m - 2$. After $2m - 2$, the Fourier series coefficients from f are fully truncated to 0.

Proof. Recalling that $\hat{b}_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} b(x)dx$, we have that $\int_{-\pi}^{\pi} \frac{1}{2\pi\hat{b}_0} b(u)du = 1$, and thus

$$|\tilde{f}(x) - f(x)| \leq \int_{-\pi}^{\pi} \frac{1}{2\pi\hat{b}_0} b(u) \cdot |f(x) - f(x - u)|du.$$

By our Lipschitz assumption of f , we can bound $|f(x) - f(x - u)| \leq \lambda|u|$ and thus have:

$$\max_x |\tilde{f}(x) - f(x)| = \|\tilde{f} - f\|_\infty \leq \lambda \cdot \frac{\int_{-\pi}^{\pi} |u|b(u)du}{2\pi\hat{b}_0} = \lambda \cdot \frac{\int_0^{\pi} ub(u)du}{\int_0^{\pi} b(u)du}. \quad (15)$$

In the last equality, we use that b is symmetric about zero. We have that $2 \cdot \sin(\frac{u}{2}) \leq u \leq \pi \cdot \sin(\frac{u}{2})$ for $x \in [0, \pi]$ and thus:

$$\int_0^{\pi} ub(u)du \leq \pi^4 \int_0^{\pi} u \frac{\sin(mu/2)^4}{u^4} du = \pi^4 m^2 \int_0^{\pi m} \frac{\sin(v/2)^4}{v^3} dv \leq \pi^4 m^2 \int_0^{\infty} \frac{\sin(v/2)^4}{v^3} dv.$$

The last integral evaluates of $\frac{\ln 2}{4}$, so overall we have $\int_0^{\pi} b(u) \cdot u du \leq \frac{\pi^4 \ln 2}{4} \cdot m^2$. Moreover we can check that:

$$\int_0^{\pi} b(u)du = \pi \cdot \left(\frac{2}{3}m^3 + \frac{1}{3}m\right) \geq \frac{2\pi}{3}m^3.$$

Plugging into (15) we have that:

$$\|\tilde{f} - f\|_\infty \leq 8.06 \frac{\lambda}{m}.$$

The result follows. We note that the constant above is loose: numerical results suggest the bound can be improved to $\leq \frac{\pi}{2} \frac{\lambda}{m}$. $\square$

Theorem B.5 translates to a result for *algebraic polynomials* via a standard transformation between Fourier series and Chebyshev series, which we detail below.

B.3 Jackson's Theorem for Algebraic Polynomials

Theorem B.6. *Let $f \in \mathcal{F}([-1, 1], \mathbb{R})$ be a Lipschitz continuous function on $[-1, 1]$ with Lipschitz constant λ . I.e., $|f(x) - f(y)| \leq \lambda|x - y|$ for all x, y . For integer m , let $\hat{b}_0, \dots, \hat{b}_{2m-2}$ be the coefficients from (14). Let $c_k = \langle f, w \cdot \bar{T}_k \rangle$ be the k^{th} coefficient in f 's Chebyshev polynomial expansion, where w and $\bar{T}_k$ are as defined in Section 2. The degree $(2m - 2)$ algebraic polynomial*

$$\tilde{f}(x) = \sum_{n=0}^{2m-2} \frac{\hat{b}_k}{\hat{b}_0} c_k \cdot \bar{T}_k(x)$$

satisfies $\|\tilde{f} - f\|_\infty \leq 9 \frac{\lambda}{m}$.

Proof. To translate from the trigonometric case to the algebraic setting, we will use the identity that for all k ,

$$T_k(\cos \theta) = \cos(k\theta). \quad (16)$$

Consider any function $r \in \mathcal{F}([-1, 1], \mathbb{R})$ with Chebyshev expansion coefficients $c_0, c_1, \dots$, where $c_k = \langle r, w \cdot \bar{T}_k \rangle$. Transform r into a periodic function as follows: let $g(\theta) = r(\cos \theta)$ for $\theta \in [-\pi, 0]$

and let $h(\theta) = g(-|\theta|)$ for $\theta \in [-\pi, \pi]$. The function $h(\theta)$ is periodic, and also even, so its Fourier series has all coefficients $\beta_1, \beta_2, \dots$ equal to 0. We thus have that

$$h(\theta) = \sum_{n=0}^{\infty} \alpha_n \cos(n\theta),$$

where

$$\alpha_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} h(\theta) \cos(k\theta) d\theta = \frac{1}{\pi} \int_{-\pi}^0 g(\theta) \cos(k\theta) d\theta$$

and, for $n > 0$,

$$\alpha_k = \frac{1}{\pi} \int_{-\pi}^{\pi} h(\theta) \cos(k\theta) d\theta = \frac{2}{\pi} \int_{-\pi}^0 g(\theta) \cos(k\theta) d\theta.$$

Using (16) and the fact that $\frac{d}{dx} \cos^{-1}(x) = \frac{1}{\sqrt{1-x^2}}$, we have:

$$\int_{-\pi}^0 g(\theta) \cos(k\theta) d\theta = \int_{-1}^1 r(x) T_k(x) \frac{1}{\sqrt{1-x^2}} dx.$$

We conclude that the Chebyshev coefficients of r are precisely a scaling of the Fourier coefficients of h . Specifically, since $\bar{T}_0 = \sqrt{\frac{2}{\pi}} T_0$ and $\bar{T}_k = \sqrt{\frac{1}{\pi}} T_k$, we have:

$$\sqrt{\frac{2}{\pi}} c_0 = \alpha_0, \quad \sqrt{\frac{1}{\pi}} c_k = \alpha_k \text{ for } k > 0. \quad (17)$$

With this fact in hand, Theorem B.6 follows almost immediately from Theorem B.5. Specifically, given $f \in \mathcal{F}([-1, 1], \mathbb{R})$ with Chebyshev series coefficients $c_0, c_1, \dots$, we let $g(\theta) = f(\cos \theta)$ and $h(\theta) = g(-|\theta|)$. Let $\alpha_0, \alpha_1, \dots$ denote h 's non-zero Fourier coefficients. Then, let $\tilde{h}$ be the approximation to h given by Theorem B.5. $\tilde{h}$ is a $2m - 2$ degree trigonometric polynomial and is even since h is even and the bump function b is symmetric. Denote $\tilde{h}$'s non-zero Fourier coefficients by $\tilde{\alpha}_0, \dots, \tilde{\alpha}_{2m-2}$. We have that $\tilde{\alpha}_k = \frac{\hat{b}_k}{b_0} \alpha_k$ for $0 \leq k \leq 2m - 2$. Finally, let $\tilde{f} \in \mathcal{F}([-1, 1], \mathbb{R})$ be defined by $\tilde{f}(\cos(\theta)) = h(-\theta)$. By (17), we have that $\tilde{f}$ is a degree $2m - 2$ polynomial and its Chebyshev series coefficients $\tilde{c}_0, \dots, \tilde{c}_{2m-2}$ are exactly equal to $\frac{\hat{b}_k}{b_0} c_k$.

Moreover, we have $\|f - \tilde{f}\|_{\infty} = \max_{x \in [-1, 1]} |f(x) - \tilde{f}(x)| = \max_x |h(x) - \tilde{h}(x)|$. By Theorem B.5 we have $\max_x |h(x) - \tilde{h}(x)| < 9 \frac{\lambda}{m}$, so we conclude that $\|f - \tilde{f}\|_{\infty} < 9 \frac{\lambda}{m}$. $\square$

In addition to the main result of Theorem B.6, our SDE algorithm also require an additional property of the damped Chebyshev approximation $\tilde{f}$:

Lemma B.7. *For any non-negative function $f \in \mathcal{F}([-1, 1], \mathbb{R})$ (not necessarily Lipschitz), let $\tilde{f}$ be as in Theorem B.6. We have that $\tilde{f}$ is also non-negative on $[-1, 1]$.*

Proof. Let $h(\theta)$ and $\tilde{h}(\theta)$ be the 2π perioduc functions as in the proof of Theorem B.6. I.e., $h(\theta) = g(-|\theta|)$ where $g(\theta) = f(\cos \theta)$ and $\tilde{h}$ is the truncated, Jackson-damped approximation to h from Theorem B.5. If f is non-negative, then so is h , and since $\tilde{h}$ is the convolution of h with a non-negative function, it is non-negative as well. Finally, since $\tilde{f}(\cos(\theta)) = h(-\theta)$, we conclude that $\tilde{f}(x) \geq 0$ for $x \in [-1, 1]$. $\square$